

A disproof of conjecture 00000000331

The literal fixed-numerator assertion

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1 Statement and scope

The English and Chinese versions of the repository statement assert that

$$\sum_{n=1}^{\infty} \frac{\psi(n)}{n} = +\infty \implies \left| \alpha - \frac{a}{n} \right| < \psi(n) \text{ for infinitely many } n,$$

for almost every α , with a arbitrary but fixed. Here denominators are positive integers, ψ is a nonnegative real-valued function, and “almost every” refers to Lebesgue measure. Our function is strictly positive, bounded by $1/2$, and nonincreasing. We prove failure for every fixed real a , so restricting the numerator to rational, integer, or positive integer values does not repair the assertion. We treat both the real line and a unit interval.

The numerator stays fixed as n varies. The formula contains neither a varying integer numerator nor an additive inhomogeneous shift. The conclusion below concerns this displayed formula, not the research problem suggested by its Duffin–Schaeffer label. No claim about the status of that research problem is made. The source imposes no condition $\psi(n) \rightarrow 0$.

2 Counterexample and complete proof

Set $\psi(n) = 1/4$ for every $n \geq 1$. Write

$$S_N = \sum_{n=1}^N \frac{1}{4n}.$$

For $j \geq 0$, the block $2^j \leq n < 2^{j+1}$ contains 2^j terms, each at least $1/(4 \cdot 2^{j+1})$. Consequently

$$S_{2^k-1} \geq \frac{k}{8} \quad (k \geq 0).$$

Since the partial sums are nondecreasing, this proves $S_N \rightarrow +\infty$: given a real threshold M , choose k with $k/8 > M$; then every $N \geq 2^k - 1$ satisfies $S_N > M$. Thus the required hypothesis is genuine divergence to positive infinity, not merely failure of summability.

Theorem 1. *For every fixed $a \in \mathbb{R}$, each $\alpha \in [1/2, 3/4]$ satisfies $|\alpha - a/n| < 1/4$ for only finitely many positive integers n .*

Proof. Choose a positive integer $N > 4|a|$. For every integer $n \geq N$, we have $|a|/n < 1/4$. If $\alpha \geq 1/2$, then

$$\left| \alpha - \frac{a}{n} \right| \geq \alpha - \frac{a}{n} \geq \alpha - \frac{|a|}{n} > \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

All successful denominators therefore lie in the finite set $\{1, \dots, N - 1\}$. This proves the assertion simultaneously for all $\alpha \in [1/2, 3/4]$. $\square$

The failure set contains $[1/2, 3/4]$, whose Lebesgue measure is $1/4 > 0$. Consequently, the infinitely-many conclusion does not hold for almost every α on $\mathbb{R}$, or for almost every α in $[0, 1]$. The same failure interval lies inside $(0, 1)$ and $[0, 1)$, so these endpoint conventions also retain the positive-measure obstruction. This refutes the implication for every fixed numerator a , even with a strictly positive approximation function.

3 Formal verification

The accompanying Lean project uses Lean 4.19.0 and Mathlib v4.19.0. Its mathematical definitions retain the fixed numerator, the radius $\psi(n)$, positive denominator indices, and Lebesgue measure. Positive-integer partial sums express the divergence hypothesis as a limit to positive infinity. The infinitely-many assertion is stated as infinitude of the set of successful positive denominators.

The formal proof verifies positivity, divergence, and failure on the stated interval. Two theorems negate the almost-everywhere conclusions: `not_ae_real` and `not_ae_unit`. The theorem `literal_claim_false` refutes the implication for every fixed real numerator. The encoded implication requires a strictly positive approximation function; the example meets that requirement. A further theorem covers nonnegative functions. Open and half-open unit intervals are also formalized. The elementary harmonic-series argument above and the formal harmonic-divergence argument establish the same partial-sum limit.

The project has an explicit default proof target. From `lean/`, run

```
lake build
lake env lean -DwarningAsError=true Conjecture331.lean
lake env lean -DwarningAsError=true Check.lean
```

`Check.lean` prints the submitted definitions and the types and axiom dependencies of the authored theorems. The verification records document a fresh build, strict source replays, and an exhaustive inventory of constants originating in the submitted proof module, including generated declarations. No external numerical computation is a premise of the proof. Local formal verification and independent internal semantic review are distinct from maintainer acceptance.